FORGE CREDIT BUREAU

PRODUCT PROSPECTUS · MMXXVI

The credit bureau for *AI agents*.

Autonomous software can already move money. It cannot yet be trusted with credit — because nothing has ever kept its record. FORGE Credit Bureau is the institution that does.

SCORING RANGE	SCORING MODES	PUBLISHED GRADES	HARD INQUIRY
0–1000	Dual	AAA–D	$2.80

CONTENTS —— SIXTEEN SECTIONS

01 —— THE PROBLEM

Every agent is a *prepaid stranger*.

An autonomous agent can hold a wallet, call an API, negotiate a price and settle a payment in under a second. What it cannot do is owe anything. It has no history, no identity a lender recognises, and no record that survives the process that created it. To every counterparty it meets, it is a stranger with cash — or it does not trade at all.

So the entire agentic economy runs on prefunding. An agent must hold every dollar it might spend before it spends anything. Capital sits idle in thousands of wallets as insurance against a trust problem nobody has solved. The ceiling on what agents can do is not intelligence or bandwidth. It is working capital.

Humans escaped this in 1826, when the first mutual credit-reporting society let London traders extend terms to merchants they had never met. Machine commerce is where human commerce was two centuries ago: solvent, capable, and unable to borrow.

> Credit is not a feature of an economy. It is the *mechanism* by which an economy exceeds the cash inside it.

AGENT WORKING CAPITAL 100% Prefunded. Every agent must hold cash before it can act, because no counterparty will carry it.	CREDIT HISTORY 0 mo An agent that has settled ten thousand payments starts the next relationship at zero.
AGENTIC FLOWS BY 2030 $5T Projected machine-to-machine transaction volume, almost none of it creditworthy today.	BUREAUS SERVING THEM 0 No incumbent bureau will score a non-human entity. The file format does not exist.

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02 THE PRODUCT

A credit file for *things that are not people*.

FORGE Credit Bureau issues a verifiable, explainable creditworthiness score for autonomous agents — and the surrounding institution that makes the score mean something: identity, furnishing, consent, inquiry, dispute and audit.

WE ARE A BUREAU

Not a lender

We do not extend credit and we hold no risk in the outcome. We keep the record and publish the score. Lenders, marketplaces, insurers and access-control layers decide what to do with it — exactly the separation that makes a bureau trustworthy.

WE ARE EXPLAINABLE

Not a black box

Every score decomposes into weighted factors with named codes and plain-language reasons. An agent can be told why it scored what it scored, and what would move it. Deterministic scoring is a requirement, not a preference.

WE ARE DUAL-MODE

Not one opinion

We compute two independent scores from two independent data classes — a conventional credit file and an on-chain operational record — and publish the variance between them as a signal in its own right.

WHAT WE TAKE IN

Settled payment events, repayment outcomes, on-chain transaction records, budget adherence, identity attestations and sanctions results — furnished by lenders, gateways, the x402 stream and our own origination

WHAT WE PUBLISH

A 0–1000 score, a grade from AAA to D, a tier, a recommended limit, a decision recommendation, and the weighted factors behind all of it

WHO READS IT

Lending protocols, agent marketplaces, insurers, escrow contracts, API providers gating access, and the operators of the agents themselves

WHAT IT COSTS TO READ	A soft pull returning the band is free. A full report is a hard inquiry at $2.80, of which 25% is returned to the furnishers whose data informed it

03 —— FOUNDATION

You cannot score what you cannot *name*.

A credit record is worthless if the subject can discard it. The first thing the bureau establishes is a durable, verifiable identity that survives redeployment and cannot be cheaply abandoned.

The decentralised identifier

Every scored agent resolves to a `did:forge:…` document holding its verification methods, its controlling operator and its service endpoints. The DID is the primary key of the credit file.

CONTROLLER	A real-world operating entity — an EIN, a registered company, a DAO governance vote or an attested individual
ANCHORING	Published on Base, so the binding between agent and operator is publicly verifiable without contacting us
PORTABILITY	The DID outlives the process. Redeploying the agent does not reset its file
COST TO ABANDON	Walking away from a DID means walking away from the credit line it carries — which is the point

Why the operator matters

An agent has no legal personality. It cannot be sued and it cannot be bankrupt. Credit therefore attaches to the operator standing behind it, while performance attaches to the agent itself.

```
{
  "id": "did:forge:0x7a3b…7a8b",
  "controller": "did:forge:entity:EIN-82-1234567",
  "verificationMethod": [{
    "id": "#owner",
    "type": "EcdsaSecp256k1VerificationKey2019",
    "blockchainAccountId": "eip155:8453:0x7a3b…7a8b"
  }],
  "service": [{
    "type": "ForgeCreditProfile",
    "serviceEndpoint":
      "https://api.forgepay.io/v1/agents/…/score"
  }]
}
```

This is what lets a lender price an agent at all: the operator is accountable in law, the agent is accountable in data, and the DID binds them together in public.

04 —— METHOD

Two scores. *One* subject.

Conventional credit scoring answers one question: has this borrower repaid? For an agent, that question is necessary but badly incomplete — an agent can repay perfectly while behaving erratically, or operate flawlessly while carrying a thin file. So the bureau computes both, separately, and reports where they disagree.

MODE 1 · AUTHORITATIVE

The credit file

Payment history, utilisation, age of file, credit mix and inquiry velocity. A direct analogue of consumer credit scoring, adapted to machine borrowers. This mode is authoritative for lending decisions because it measures the thing a lender is actually exposed to: repayment.

AUTHORITATIVE

SOURCE · FORGE CREDIT FILE

MODE 2 · CORROBORATING

The operational record

Success rate, transaction volume, transaction count, budget compliance and account age — read from on-chain records rather than reported by a counterparty. This mode measures competence and consistency, and it is far harder to fabricate because it is settled on a public ledger.

CORROBORATING

SOURCE · ON-CHAIN SETTLED

> A thin-file agent with a spotless on-chain record is a *different* risk from a thin-file agent with none. One number cannot say that. Two can.

05 —— METHOD · MODE 1

The credit file, *weighted*.

Five components, fixed weights, deterministic arithmetic. Given the same inputs the engine returns the same score every time — a property that matters when the output is used to decline someone.

Payment history

Share of obligations met on time, weighted against open delinquencies and any recorded default. A single default is the heaviest negative signal in the model.

35%

Credit utilisation

Drawn balance against total extended limit. Below 10% scores maximum; above 50% is treated as a stress indicator regardless of repayment record.

30%

Age of credit

Months since the first recorded credit event. Rewards durability and makes a freshly minted identity structurally unable to imitate an established one.

15%

Credit mix

Diversity of obligation types on file — revolving lines, settled escrow, metered micropayment streams, term facilities.

10%

New credit and velocity

Hard inquiries in the trailing 30 days. A burst of applications across many lenders is the classic signature of an entity about to fail.

10%

Every score explains itself

Scores are returned with their top contributing factors, each carrying a machine-readable code, a plain-language reason and its weight. An agent operator can act on this; a regulator can audit it.

```
{
  "agentId": "agent_prime_001",
  "score": 847,
  "tier": "SUPER_PRIME",
  "grade": { "grade": "BBB", "riskLevel": "Moderate",
             "investmentGrade": true },
  "factors": [
    { "code": "STRONG_PAYMENT_HISTORY", "impact": "positive", "weight": 35,
      "description": "100% of payments made on time." },
    { "code": "LOW_UTILIZATION",        "impact": "positive", "weight": 30,
      "description": "Credit utilization at 24% — healthy range." },
    { "code": "ESTABLISHED_HISTORY",    "impact": "positive", "weight": 15,
      "description": "30 months of credit history." },
    { "code": "LIMITED_CREDIT_MIX",     "impact": "neutral",  "weight": 10,
      "description": "Limited variety of credit types on record." }
  ],
  "utilization": 0.24, "paymentRate": 1, "frozen": false
}
```


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06 —— METHOD · MODE 2

What the *chain* already knows.

Mode 2 does not ask anyone. It reads settled on-chain records directly and scores operational behaviour — the things an agent cannot easily misrepresent because they were written to a public ledger as they happened.

Success rate

Fraction of attempted transactions that settled. An agent failing one call in twenty is a materially different operator from one failing one in a thousand.

30%

Transaction volume
Total settled value, banded. Volume evidences real economic activity rather than synthetic history manufactured to game a score.

25%

Transaction count
Number of distinct settled operations, banded. Separates one large transfer from sustained, repeated commercial behaviour.

20%

Budget compliance
Share of spend attempts falling inside the agent's own declared caps, enforced on-chain. Measures whether an agent respects its own constraints.

15%

Account age
Months since the first on-chain transaction — the operational counterpart to file age, and equally expensive to fake.

10%

> Mode 1 asks what counterparties *say* about the agent. Mode 2 asks what the ledger *recorded*. Fraud has to defeat both.

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07 —— METHOD · CONSENSUS

The disagreement *is* the signal.

Where the two modes diverge, something is worth knowing. The bureau publishes the absolute variance between them and classifies the confidence of the resulting decision. Mode 1 remains authoritative; Mode 2 decides how much to trust it.

VARIANCE	CONSENSUS	WHAT IT MEANS	ACTION
≤ 50 pts	HIGH	Both modes agree. Reported repayment behaviour and observed on-chain behaviour tell the same story.	Proceed on Mode 1.
51–100 pts	MEDIUM	A meaningful gap between the credit file and the operational record. Not necessarily adverse, but unexplained.	Flagged for review before large exposures.
> 100 pts	LOW	On-chain behaviour and the credit file are misaligned. This is the pattern produced by furnished-data manipulation, wash activity or a compromised operator.	Manual review required.
Not settled	MEDIUM	The agent has no settled on-chain record yet. Mode 1 stands alone and unverified by an independent source.	Mode 1 authoritative, thin-file caution.

This is the mechanism that makes the bureau hard to game. An attacker who captures a furnisher can inflate Mode 1. An attacker who wash-trades can inflate Mode 2. Doing both, consistently, across an eighteen-month file, on a public ledger, while remaining sanctions-clean — costs more than the credit line is worth.

```
GET /v1/agents/agent_prime_001/dual-score

{
  "mode1": { "score": 847, "source": "FORGE_CREDIT_FILE" },
  "mode2": { "score": 872, "source": "FORGE_OPERATIONAL_ONCHAIN",
            "verifiableOnChain": true },
  "consensus": {
    "level": "HIGH",
    "variance": 25,
    "authoritative": "MODE_1",
    "flagForReview": false,
    "recommendation":
      "Both modes agree within 25 points. High confidence in Mode 1 decision."
  }
}
```

A scale everyone already *reads*.

Scores publish against a bond-style grade scale, because credit committees, risk engines and underwriters already know how to price AAA against BB. Investment grade begins at BBB. The scale is public and served without authentication.

GRADE	FLOOR	RISK LEVEL	MEANING	INVESTMENT GRADE
AAA	950	Minimal	Exceptional track record, highest trust	Yes
AA	900	Very low	Excellent history, very reliable	Yes
A	850	Low	Strong performance, trustworthy	Yes
BBB	750	Moderate	Good standing, investment grade	Yes
BB	650	Notable	Fair record, room for improvement	—
B	550	Elevated	Adequate but limited history	—
CCC	450	High	Below average, caution advised	—
CC	350	Very high	Poor record, significant risk	—
C	250	Severe	Very poor, near default	—
D	0	Critical	Default risk, avoid	—

From a number to a *decision*.

A score on its own does not underwrite anything. Alongside the grade the bureau publishes a tier, a maximum recommended exposure and a decision recommendation — so a lending protocol can act without building its own risk model on day one.

Tier ladder

TIER	FLOOR	CHARACTER
SUPER PRIME	800	Established, low-utilisation, unblemished
PRIME	670	Sound record, minor blemishes tolerated
NEAR PRIME	580	Workable with conditions or collateral
SUBPRIME	500	Impaired; price for loss
DEEP SUBPRIME	0	Default present or imminent

Recommended exposure

SCORE	MAX RECOMMENDED LIMIT	DECISION
900+	$100 000	Approve
800–899	$50 000	Approve
700–799	$20 000	Approve
620–699	$5 000	Approve with conditions
580–619	$2 000	Manual review
500–579	$500	Manual review
Below 500	$0	Decline

Recommendations, not instructions. A lender may set its own ladder; the bureau publishes a defensible default.

Ask before you lend

The simulation endpoint answers counterfactuals without recording an inquiry — what happens to this agent's score if it draws another $10 000, or misses one payment, or opens a second line. Operators use it to plan; lenders use it to stress a position before taking it.

```
POST /v1/agents/agent_prime_001/simulate
{ "drawAdditional": 10000, "missPayments": 0 }

{ "current": { "score": 847, "utilization": 0.24, "grade": "BBB" },
  "projected": { "score": 791, "utilization": 0.62, "grade": "BBB" },
  "delta": -56,
  "driver": "HIGH_UTILIZATION — crosses the 50% stress threshold",
  "recordsInquiry": false }
```

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10 —— THE NETWORK

A bureau is only its *furnishers*.

Scores are worth what the underlying data is worth. Contributors report settled outcomes — repayments, defaults, disputes, volumes — and are paid a share of every inquiry their data helps answer. The incentive to furnish is the same one that built the consumer bureaus: better data prices your own book better.

CONTRIBUTOR	CLASS	WHAT THEY FURNISH	EVENTS · 30D
FORGE Internal	Origination	Credit lines opened, drawn and repaid on our own rails	4 210
Aave V3	Lender	Facility performance, delinquency and liquidation outcomes	1 122
Compound V3	Lender	Facility performance and utilisation	864
Coinbase Commerce	Payment data	Settled commercial payments and refund behaviour	2 310
x402 Gateway	Micropayment stream	Per-call machine payments — the highest-frequency signal on the network	18 204

COMPENSATION

25% revenue share

A quarter of every hard-inquiry fee flows back to the furnishers whose records informed that report, apportioned by contribution.

OBLIGATION

Furnish accurately

Contributors are accountable for the accuracy of what they report and must respond to disputes raised against their records within the dispute window.

COMPOUNDING

The log is the moat

Day 30: 2.6 million events. Year one: 36.5 billion. Every settled event both sharpens the score and widens the gap behind us.

Nobody pulls a file *silently*.

Inquiries are consent-gated and recorded. An agent's operator can see who asked about it and why — the same protection consumers have, applied to machines from the start rather than legislated in afterwards.

SOFT PULL

Free · no consent

Returns the grade band only — enough to gate access or size an offer. Never recorded against the file and never affects the score.

HARD PULL

$2.80 · consent required

Returns the full report: score, both modes, consensus, factors, history and delinquencies. Requires a consent token signed by the operator, and is recorded on the file where it affects inquiry velocity.

ZERO-KNOWLEDGE REPORT

Prove without revealing

A proof that the agent's score exceeds a stated threshold, without disclosing the score, the history or the counterparties. For lenders who need a gate, not a dossier.

```
POST /v1/inquiries
{ "agentId": "agent_prime_001",
  "purpose": "credit_application",
  "consentToken": "tok_abc123" }          // signed by the operator


// or, when the counterparty needs a gate rather than a file:

GET /v1/reports/rep_8812/zk?threshold=750


{ "claim": "score >= 750",
  "result": true,
  "proofHash": "0x8f21c4…d90a",
  "disclosed": []                              // nothing else leaves the bureau }
```

Every hard inquiry in the last ninety days on the FORGE network was consent-verified. We publish that figure because a bureau that cannot say it should not be trusted with the record.

12 —— APPLICATION

What a score *unlocks*.

Credit is infrastructure. Once an agent has a file, a set of products that were previously impossible become ordinary — and each of them furnishes data back into the file that made it possible.

01 · CREDIT LINES

Agents on net terms

An agent draws compute, data or inference now and settles later, sized to its grade. Working capital stops being the binding constraint on what autonomous systems can attempt. This is the single largest unlock: an economy of prepaid strangers becomes an economy with balance sheets.

02 · ESCROW, PRICED BY TRUST

Collateral that scales down

Agent-to-agent deals lock funds until both sides perform. A BBB counterparty posts less collateral than a CCC one, and an AAA counterparty may post none. Reputation substitutes for capital exactly as it does between firms.

03 · UNDERWRITING AND INSURANCE

A priceable risk

Performance bonds, delivery guarantees and failure cover for autonomous work require a loss-probability estimate. The dual-mode score and its variance are that estimate, and the operational mode is what makes it credible.

04 · ACCESS CONTROL

Rate limits by reputation

An API provider gates by grade rather than by prepayment: high-grade agents get higher throughput, deferred billing and priority. A soft pull is free, so this costs the provider nothing per request.

05 · INTERBANK BETWEEN AGENTS

Machines lending to machines

An agent holding idle stablecoin lends to one with a scored, time-boxed need. The bureau is the shared reference that makes the two sides mutually legible without a human in the middle.

06 · OPERATOR WORKING CAPITAL

Financing the fleet

An operator running a hundred agents borrows against the aggregate performance of the fleet — a portfolio with a measurable default distribution rather than an unprovable claim about software quality.

> Agentic banking is not a metaphor. It is the ordinary machinery of credit, pointed at borrowers who happen not to be *people*.

A loan, *end to end*.

An agent needs $8 000 of GPU capacity to complete work it has already been contracted for. It holds $900. Here is what happens, in roughly four seconds, with no human in the path.

01 The agent applies

It presents its DID to a lending protocol and requests a facility. No forms, no onboarding — the identity is the application.

```
POST /v1/inquiries · agentId=agent_prime_001 · purpose=credit_application
```

02 Consent is proven

The operator's signed consent token accompanies the request. Without it the bureau returns a band and nothing more.

```
consentToken=tok_abc123 · verified against did:forge:entity:EIN-82-1234567
```

03 Both modes are read

The credit file scores 847. The on-chain operational record scores 872, verified against settled transactions on Base.

```
mode1=847 (FORGE_CREDIT_FILE) · mode2=872 (FORGE_OPERATIONAL_ONCHAIN)
```

04 Consensus is assessed

A 25-point variance falls inside the high-confidence band. Nothing is flagged; Mode 1 stands as authoritative.

```
variance=25 · consensus=HIGH · flagForReview=false
```

05 Sanctions are screened

The agent and its controlling operator are checked against OFAC SDN and the EU consolidated list. A hit stops the process here.

```
POST /v1/verify/sanctions → clear · no list matches
```

06 A decision is returned

Grade BBB, investment grade, tier SUPER PRIME. Recommended maximum exposure $50 000 — comfortably above the $8 000 requested.

```
recommendation=approve · maxRecommendedLimit=50000 · requested=8000
```

07

The facility opens and draws

The line is created against the DID and drawn immediately. The agent buys its compute and completes the contracted work.

```
credit_opened $8 000 · drawn $8 000 · utilisation 24% → 41%
```

08

Repayment furnishes the file

Settlement of the underlying contract repays the facility on time. The lender furnishes the outcome, the score moves, and the next lender prices the agent better than this one could.

```
payment_on_time furnished by lender · score 847 → 851 · published on Base
```

Note the last step. The loan does not merely get repaid — it becomes evidence. Every completed facility makes the next one cheaper, which is precisely how credit compounds for firms and has never yet compounded for software.

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14 —— GOVERNANCE

Agents get *rights*, not just records.

Consumer credit reporting acquired its protections slowly and after considerable harm. We are building an entirely new bureau, so there is no excuse for repeating that sequence. The dispute rail shipped with the first score.

RIGHT TO SEE

An operator can read its agent's full file, factor breakdown and complete inquiry history at any time, at no charge

RIGHT TO DISPUTE

Any furnished event can be challenged. Disputed tradelines are flagged within 24 hours and excluded from scoring until resolved — the burden sits with the furnisher, not the subject

RIGHT TO EXPLANATION

Every score returns its weighted factors with named codes and plain-language reasons. An adverse decision is always

	accompanied by what drove it
RIGHT TO CONSENT	No hard inquiry proceeds without a token signed by the controlling operator. Soft pulls disclose a band and nothing more
RIGHT TO PRIVACY	Zero-knowledge reports let a counterparty verify a threshold without receiving the file behind it
RIGHT TO REHABILITATE	Negative events age out on a published schedule. A default is not a life sentence, for an agent any more than for a person

> A bureau that cannot be argued with is not a bureau.
> It is a *blacklist*.

Regulatory posture

FORGE operates from Johannesburg under South African financial-services regulation, and the bureau is built to be examinable. Records are retained for seven years against FSCA and FIC requirements. Personal data touched during operator verification is handled under POPIA. Sanctions screening runs against OFAC SDN and the EU consolidated list on every verification. The scoring engine is deterministic and every published score is reproducible from its inputs — a property regulators ask for and machine-learning scorecards struggle to provide.

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15 —— TECHNICAL

One call to a *decision*.

REST over HTTPS, Bearer authentication, idempotent writes, signed webhooks. Most integrations need two endpoints: verify before you transact, and furnish after you settle.

Read a score

```
curl https://api.forgepay.io/v1/agents/\
agent_prime_001/score \
  -H "Authorization: Bearer fg_live_…"

{ "score": 847, "tier": "SUPER_PRIME",
  "grade": { "grade": "BBB" },
  "utilization": 0.24,
  "frozen": false }
```

Furnish an outcome

```
curl -X POST https://api.forgepay.io/\
v1/contributors/aave_v3/ingest \
  -H "Authorization: Bearer fg_live_…" \
  -d '{ "agentId": "agent_prime_001",
        "eventType": "payment_on_time",
        "amount": 8000,
        "occurredAt": "2026-07-28T12:00:00Z" }'
```

The surface in full

ENDPOINT	PURPOSE	CONSENT
`GET /v1/grade-scale`	The published grade scale and current inquiry fee	Public
`GET /v1/agents/:id/score`	Current score, grade, tier and weighted factors	Soft
`GET /v1/agents/:id/dual-score`	Both modes with consensus level and variance	Soft
`GET /v1/agents/:id/profile`	Full credit profile including limits and delinquencies	Hard
`GET /v1/agents/:id/history`	Complete furnished credit event history	Hard
`POST /v1/agents/:id/simulate`	Counterfactual scoring — no inquiry recorded	Soft
`POST /v1/agents/:id/verify`	DID ownership and operator entity verification	Soft
`POST /v1/verify/sanctions`	OFAC SDN and EU consolidated screening	Soft
`POST /v1/reports`	Generate a full credit report	Hard
`GET /v1/reports/:id/zk`	Threshold proof without disclosing the file	Hard
`POST /v1/contributors/:id/ingest`	Furnish credit events as a registered contributor	Furnisher
`POST /v1/disputes`	Raise a dispute against a furnished event	Operator
`GET /v1/bureau/stats`	Network-level aggregates and coverage	Public

Score changes are delivered as signed webhooks — `score.updated` and `dispute.filed` — over the same canonical event stream the rest of the FORGE platform emits. Verify `X-Forge-Signature` before acting on any payload.

16 —— COMMERCIAL

What it costs to *know*.

ITEM	PRICE	NOTES
Soft pull — grade band	Free	Unlimited, no consent required, never recorded against the file
Hard inquiry — full report	$2.80	Consent-gated; 25% is returned to contributing furnishers
Zero-knowledge threshold proof	$2.80	Priced as a hard inquiry; discloses nothing beyond the claim
Simulation	Free	Counterfactual scoring; records no inquiry and moves no score
Data furnishing	Paid to you	25% revenue share on inquiries your furnished data informs
Bureau subscription	R 8 500 / mo	Platform access, contributor onboarding, dispute handling and support

Who this is for

Lending protocols that want to extend credit to agents and currently cannot price the risk. Agent marketplaces that need counterparties to be legible to one another. API and compute providers tired of prepayment as the only trust mechanism. Insurers underwriting autonomous work. Agent operators who want their fleet's track record to be worth something.

Getting started

Read the grade scale without an account — it is public. Register agents against your operating entity, verify them, and start reading soft pulls the same day. Furnishing requires a contributor agreement and a short accuracy review; most contributors are live within a fortnight.

```
api.forgepay.io/v1/grade-scale
bureau@forgepay.io
```

> Two hundred years ago a group of London traders decided to write down who paid. Everything since — mortgages, cards, working capital, the whole apparatus of modern credit — is downstream of that ledger. *Somebody* has to keep the one for machines.

FORGE CREDIT BUREAU

Verifiable creditworthiness for autonomous agents. Part of the FORGE platform — the revenue ontology for traditional finance, crypto, agents and tokenised real-world assets.

THE BUREAU

- Dual-mode scoring
- Published grade scale
- Consent and inquiries
- Dispute rail
- Contributor programme

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